

GeoNLI : Natural Language Interpretation of Satellite Imagery

Abstract—The rapid expansion of Earth observation capabilities has produced an unprecedented volume of high-resolution satellite imagery. Extracting actionable insights from this data typically requires specialized GIS expertise, creating a significant barrier for wider scientific and operational use. This report introduces a natural language-driven prototype that enables intuitive interaction with processed satellite imagery, allowing non-expert users to obtain meaningful interpretations without technical training. The system combines large language models with vision language models to support three core tasks: captioning, which generates concise descriptions of scene-level content; grounding, which localizes queried objects via oriented bounding boxes; and visual question answering, which encompasses binary, semantic, and numeric queries.

I. Introduction

The democratization of space has created a massive influx of high-resolution Earth observation data. While this imagery captures critical details about our planet, from urban growth to vegetation health, there is a major barrier to using it: the “semantic gap.” The raw pixels are available, but understanding them currently requires expert knowledge in Geographic Information Systems (GIS). This prevents non-experts, such as disaster response teams or urban planners, from simply asking questions about a satellite image and getting a clear answer.

The core problem is that satellite imagery looks very different from the normal photos used to train standard AI models.

- **Top-Down View:** Satellites look straight down (nadir view), making objects appear flat and removing the depth cues present in natural images.
- **Scale and Density:** Objects such as cars or aircraft often occupy only a handful of pixels and appear in dense clusters.
- **Visual Ambiguity:** Shadows, roof patterns, and man-made structures can resemble each other, creating confusion even for advanced perception models.



Fig. 1: A satellite scene illustrating the core challenges: extreme object density, long shadows obscuring features, and tiny objects that are difficult to resolve.

As shown in Fig. 1, these issues confuse standard models. When a generic AI looks at a satellite image, it often treats the image as a loose collection of visual tokens rather than a spatially structured scene. Recent research confirms that this lack of spatial grounding leads models to miss small objects or misunderstand relationships like “next to” or “aligned with” [1], [10]. As a result, off-the-shelf models frequently generate overly long captions, hallucinate objects, or misinterpret simple counting tasks.

To bridge this gap, we present a system specifically adapted for remote sensing. Our prototype enables three key capabilities: (1) concise scene-level captioning, (2) spatial grounding of user-specified objects, and (3) visual question answering covering binary, semantic, and numeric queries. Through domain-specific fine-tuning, the system develops a more robust understanding of satellite perspectives, small-object structures, and context-dependent visual clues.

To provide a high-level view of the system architecture, Fig. ?? outlines the end-to-end GeoNLI inference workflow. The pipeline consists of seven stages, beginning with environment initialization and model loading, followed by image and metadata ingestion, query dispatch, and task-specific reasoning. These reasoning modules include both generative

components (captioning and instruction following) and analytic components (object localization, spatial reasoning, and small-object analysis). The final stage formats the output into a structured, user-ready response. Although later sections discuss each task in detail, this workflow diagram establishes the overall processing logic through which the system converts raw satellite imagery into interpretable natural-language insights.

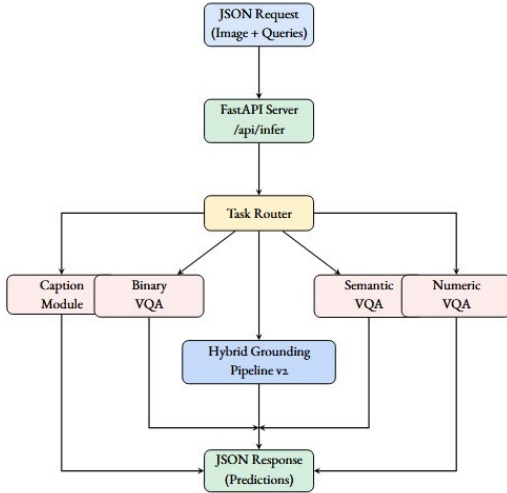


Fig. 2: Overview of the unified GeoNLI inference workflow.

II. Related Work

Current research highlights that while general purpose AI models are getting smarter, they still struggle with the unique demands of satellite imagery. This section reviews why these struggles happen and how recent papers propose fixing them.

A. Why Models Struggle with Space and Position

Even powerful Vision Language Models (VLMs) often fail at basic spatial reasoning. Recent benchmarks show that while a model might recognize a "plane," it often cannot tell if the plane is "left of" or "facing" the terminal [2]. Why is this hard? Studies suggest that standard models prioritize what an object is over where it is. The visual signals for "identity" are so strong that they drown out the subtle signals for "location" [3]. This leads to "shortcut learning," where the model guesses based on probability rather than actually looking at the spatial layout [6].

B. Seeing the Small Details

Satellite imagery is massive, but the objects of interest are often tiny. Standard models frequently miss these small details or fail to notice the "odd one out" in a group of similar objects [11]. However, new research offers solutions. One approach is "looking shallower"—using the earlier layers of the neural network, which contain more raw visual detail, rather than just the deep layers that focus on abstract concepts [8]. Other techniques involve "token packing," which compresses visual data more efficiently to handle high-resolution images without losing fine details [7], [9]. Some work even shows that models implicitly "know" where small objects are (their internal attention highlights them) even if their final text answer is wrong [5].

C. Teaching Models to Follow Instructions

For a model to be useful, it must follow complex instructions without getting confused. The "Learning to Instruct" (L2T) framework suggests that we need to balance teaching the model to follow orders with teaching it to actually understand the image content, preventing it from just memorizing prompt patterns [4]. Other frameworks, like Prism [12], suggest separating the "seeing" (perception) part of the model from the "thinking" (reasoning) part, which is helpful for remote sensing where perception is unusually difficult. Techniques like Supervised Embedding Alignment (SEA) help by aligning the visual representation of satellite objects directly with their text descriptions [13].

D. Adapting to Remote Sensing

Finally, specific adaptations for satellite data have proven effective. "Structured attention" helps the model process the complex, multi-level layouts of cities [14]. Similarly, Label-Aware Multi-level Attention (LAM) forces the model to pay attention to specific earth-observation tags (like "industrial area" or "dense forest") during generation, ensuring the output is technically accurate [15]. These works confirm that generic models need specialized training strategies to become reliable tools for earth observation.

III. Tasks

A. Captioning

Satellite image captioning aims to generate short, meaningful textual descriptions of overhead scenes by identifying key objects, land cover patterns, and spatial layouts.

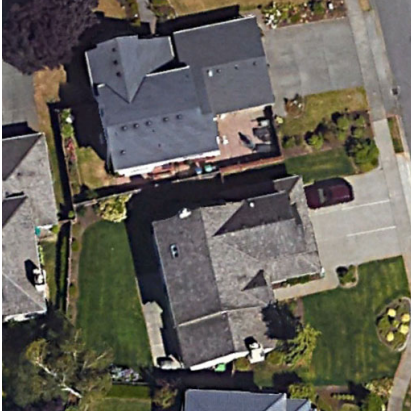


Fig. 3: Captioning example. The groundtruth and model generated captions both identify residential structures, green lawns, and the small parked vehicle, demonstrating alignment between predicted semantics and reference descriptions.

Qualitative Performance: As shown in Fig. 3, our fine tuned model demonstrates a high degree of alignment with ground truth data.

- **Ground Truth:** "The GoogleEarth image captures a residential area with houses surrounded by green lawns. Visible in the middle right part of the image is a small vehicle parked on a driveway adjacent to one of the houses."
- **Model Output:** "The high resolution image from GoogleEarth displays a residential area with houses featuring dark rooftops and large yards with green grass. A small vehicle is parked in front of one house on the bottom right corner."

The model successfully captures key semantic details ("dark rooftops", "large yards", "small vehicle") while maintaining the concise structure required for effective reporting.

1) Metric Evaluation: BERT BLEU₄

Captioning performance is evaluated using BERT BLEU₄, which measures semantic similarity between the generated caption and the reference caption using contextual BERT embeddings [16]. Let C and R be the sets of tokens in the candidate and reference captions, and let C_n and R_n denote their respective n gram sets. For each $n \in \{1, 2, 3, 4\}$, semantic recall is defined as:

$$P_n = \frac{1}{|R_n|} \sum_{r \in R_n} \max_{c \in C_n} \cos(E(c), E(r)),$$

where $E(x)$ denotes the BERT embedding of an n gram x . The overall BERT BLEU₄ score is then:

$$\text{BERT-BLEU}_4 = LP \cdot \exp \left(\frac{1}{4} \sum_{n=1}^4 \log P_n \right), \quad (1)$$

where the length penalty is defined as:

$$LP = \exp \left(-\alpha \cdot \frac{|L_C - L_R|}{L_R} \right),$$

with $\alpha = 0.5$. Equation (1) ensures that both lexical and semantic alignment are captured while penalizing captions that are too short, too long, or semantically misaligned.

a) *Interpretation:* A higher BERT-BLEU score indicates stronger semantic alignment between the generated caption and the reference caption. Because it incorporates both contextual similarity (via BERT embeddings) and structural precision (via n grams), the metric rewards models that capture the true meaning of the image rather than relying on surface level word matching.

2) Embedding Evaluation

In addition to the main BERT-BLEU₄ score, we evaluate caption quality using two complementary embedding based strategies that differ in how they encode n grams.

a) **Contextual Token Level Embedding (BERT):** The first strategy uses a full sentence BERT encoder (bert base uncased). The entire caption is passed through BERT once, and the contextualized token embeddings are extracted. For any n -gram, its vector representation is computed as the average of the corresponding token embeddings. Because BERT embeds each token in the context of the full sentence, this method captures semantic meaning that depends on surrounding words (e.g., "river bank" vs. "bank deposit"). This enables fine grained matching of phrases in a contextual setting with high efficiency.

b) **Phrase Level Sentence Embeddings (MiniLM / SBERT):** The second strategy treats every n gram as an independent phrase. Each constructed n gram string (e.g., "small vehicle", "green lawn") is encoded individually using a Sentence Transformer model (MiniLM) [17]. Unlike contextual BERT embeddings, these vectors do not depend on the surrounding sentence they reflect the inherent semantic meaning of each phrase. Although computationally more expensive (because each n -gram requires a separate forward pass), this method provides strong standalone phrase representations optimized directly for cosine similarity.

3) Difficulties

A recurring issue during experimentation was that models produced excessively long captions. These verbose descriptions often contained redundant details, lowering BERT-BLEU scores due to length mismatches. To address this, we

TABLE I: CAPTIONING PERFORMANCE USING TOKEN-LEVEL BERT EMBEDDINGS (BERT-BLEU₄)

Model	Prompt Strategy	BERT-BLEU ₄
Qwen3-VL-4B-Instruct	Standard Few-Shot	0.4333
Qwen3-VL-4B-Instruct	Metric-Aware Few-Shot	0.6120
Qwen2.5-VL-3B-Instruct	Standard Few-Shot	0.5793
Qwen2.5-VL-3B-Instruct	Metric-Aware Few-Shot	0.5676
Qwen2.5-VL-7B-Instruct	Standard Few-Shot	0.2369
Qwen2.5-VL-7B-Instruct	Metric-Aware Few-Shot	0.5866
Qwen3-VL-8B-Instruct	Standard Few-Shot	0.1221
Qwen3-VL-8B-Instruct	Metric-Aware Few-Shot	0.5874
Qwen3-VL-4B-Instruct-bnb-4bit	SFT + Metric-Aware Few-Shot	0.5874
Qwen3-VL-8B-Instruct-bnb-4bit	SFT + Metric-Aware Few-Shot	0.5874
Qwen2.5-VL-7B-Instruct-bnb-4bit	SFT + Metric-Aware Few-Shot	0.6841

TABLE II: CAPTIONING PERFORMANCE USING SENTENCE-LEVEL MINILM EMBEDDINGS

Model	Technique Applied	BERT-BLEU ₄
Qwen2.5-VL-3B	Standard Few-Shot	0.2244
Qwen2.5-VL-3B	Metric-Aware Few-Shot	0.6517
Qwen3-VL-8B	Standard Few-Shot	0.4451
Qwen3-VL-8B	Metric-Aware Few-Shot	0.7742

used a specialized prompt encouraging median length captions. This ensured more concise, task relevant descriptions.

4) Testing and Performance

All experiments were conducted on our internal remote sensing dataset, ensuring consistent evaluation across zero shot, few shot, and supervised settings. We leveraged state-of-the-art base models such as Qwen2.5-VL and Qwen3-VL [20], [22]. In the zero shot condition, models tended to produce long, unfocused captions with redundant descriptions of rooftops, vegetation, and background areas. Few shot prompting substantially improved structural alignment by exposing the model to examples of the desired caption length and style. However, the most consistent gains appeared during supervised fine tuning (SFT), where domain-specific patterns such as small object emphasis, shadow interpretation, and scene summarization were learned directly from data.

Experimental Setup: To optimize training efficiency, we fine tuned all VLMs using the Unsloth framework. We employed 4-bit quantization (QLoRA) to reduce memory overhead, targeting the vision encoder, language backbone, MLP layers, and attention modules. Initial experiments explored higher LoRA ranks (64) to increase adaptation capacity, but empirical results showed that a rank of 16 with an alpha of 16

provided the best balance between stability and overfitting. Models were trained for 3 epochs with the 8-bit AdamW optimizer at a learning rate of 2×10^{-4} , using a cosine learning rate schedule with 100 warmup steps. We used a per device batch size of 2 and 8 gradient accumulation steps for an effective batch size of 16.

B. Numeric VQA

Numeric queries require the system to extract quantitative information from a satellite image. These tasks fall into two categories: (1) **object counting** (e.g., aircraft, buildings, storage tanks) and (2) **geometric estimation** such as area, length, width, or perimeter. Counting requires strong object-level understanding, whereas geometric measurements rely on accurate spatial localization—something general-purpose VLMs struggle with in remote-sensing settings. For this reason, our numeric pipeline combines a fine-tuned VLM for counting with a SAM3-based grounding module for all non-count queries.

1) Pipeline Overview

We first fine-tune a 4-bit quantized Qwen2.5-VL-7B model on the VRSBenchmark counting subset, enabling reliable object-count predictions under occlusions, density variations, and small-object conditions typical in satellite imagery.

When a new query arrives, the VLM parses it to determine whether it is a **count query** or a **non-count geometric query**.

- **Count queries** (“How many airplanes are present?”) are routed directly to the fine-tuned VLM, which performs zero-shot inference on the image.
- **Non-count queries** (“What is the area of the left cropland patch?”, “What is the length of the runway?”) require grounding before measurement.

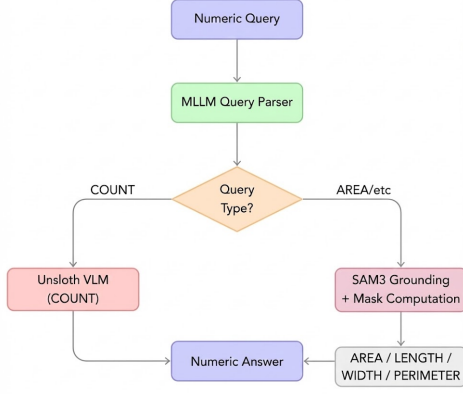


Figure 3: Hybrid routing for numeric VQA queries

Fig. 4: Hybrid routing pipeline for numeric VQA queries. Count queries are handled by a fine-tuned VLM, while non-count queries are routed through the grounding module for geometry-aware computation.

For non-count queries, the VLM further decomposes the instruction into:

- the target subject,
- any reference objects,
- spatial relations (e.g., “left of”, “top-most”),
- size or ordinal constraints,
- the physical quantity to compute (area, length, width, ratio).

These elements are passed into the grounding pipeline, which uses SAM₃ to produce candidate masks and selects the set of oriented bounding boxes that best satisfy the relational constraints. Once the correct masks are identified, geometric metrics are computed deterministically using the scene’s spatial resolution (e.g., square meters per pixel). This two-stage design—VLM parsing followed by SAM₃ localization—enables the system to handle tasks that require both semantic understanding and precise spatial measurement.

2) Metric Evaluation

Numeric accuracy is evaluated using an exponential penalty on relative error:

$$S_{\text{numeric}} = \exp \left(-\alpha \cdot \frac{|x_{\text{pred}} - x_{\text{gt}}|}{x_{\text{gt}}} \right),$$

with $\alpha = 1.0$. The metric penalizes large deviations sharply while allowing small errors common in dense or partially occluded scenes.

3) Difficulties

Counting performance is sensitive to shadows, object overlap, and dense industrial regions. Zero-shot VLM predictions

frequently underestimated object counts in such scenes. Few-shot prompting improved stability, and supervised fine tuning delivered the most consistent gains by exposing the model to characteristic remote-sensing patterns. For geometric queries, the primary difficulty was that VLMs lack reliable spatial reasoning. Integrating the SAM₃ grounding module resolved this by providing accurate masks, enabling precise measurement regardless of viewpoint or occlusion.

4) Testing and Performance

Experiments were conducted on internal RS datasets containing urban, industrial, agricultural, and airport environments. Counting accuracy improved significantly after supervised fine-tuning. A systematic sweep over LoRA ranks showed that higher ranks improved sensitivity to fine-grained visual cues. The best performance was achieved with rank 64 and $\alpha = 128$, yielding a score of 0.73 on counting queries.

TABLE III: NUMERIC QUERY PERFORMANCE ACROSS MODEL VARIANTS (ONLY ON COUNTING QUERIES)

Model	Technique Applied	Score
Qwen2-VL-7B-Instruct	Zero-shot	0.30
Qwen2-VL-7B-Instruct	2× Upscaling + Few-shot	0.40
Qwen2-VL-7B-Instruct-bnb-4bit	SFT	0.50
Qwen2-VL-7B-Instruct-bnb-4bit	SFT (Rank 16, Alpha 16)	0.61
Qwen2-VL-7B-Instruct-bnb-4bit	SFT (Rank 64, Alpha 32)	0.67
Qwen2-VL-7B-Instruct-bnb-4bit	SFT (Rank 64, Alpha 128)	0.73

C. Semantic VQA

Semantic VQA requires the model to interpret the meaning, type, or function of regions within a satellite image. Unlike numeric queries, which target measurable quantities, semantic VQA focuses on conceptual understanding such as land-use categorization, structural type identification, and functional interpretation of built environments.

1) Metric Evaluation

Semantic VQA is evaluated using the BERT-BLEU₄ measure introduced in the captioning task (Eq. 1), but applied to short free-form answers. Let C and R denote the candidate and reference responses, with C_n and R_n representing their n -gram sets. Semantic recall for each n -gram is computed as:

$$P_n = \frac{1}{|R_n|} \sum_{r \in R_n} \max_{c \in C_n} \cos(E(c), E(r)),$$

where $E(\cdot)$ denotes contextual BERT embeddings. A length penalty ensures concise outputs. The metric rewards semantic correctness even when wording differs from the reference.

2) Difficulties

General-purpose VLMs often misclassify industrial, agricultural, and peri-urban structures due to natural-image priors. Few-shot prompting provides partial correction, but domain-specific supervised fine tuning is essential for reliably aligning model outputs with remote-sensing semantics.

3) Testing and Performance

Zero-shot predictions showed moderate accuracy but frequent misinterpretations of rooftops, cropland patterns, and storage facilities. Few-shot prompting improved functional grounding, but the largest gains came from fine tuning on 5,000 RS semantic samples. The 4-bit Qwen2.5-VL-7B-Instruct model achieved a BERT-BLEU₄ score of **0.85**, demonstrating the importance of curated RS instruction data for semantic understanding.

4) Experimental Setup

Semantic VQA is implemented using the Unsloth FastVisionModel framework with a 4-bit Qwen2.5-VL-7B-Instruct backbone and LoRA adapters trained for semantic reasoning. The system loads adapters automatically, applies Unsloth’s inference optimizations, and processes each query using a two-part prompt: a system instruction enforcing concise answers and a user instruction attaching the image. Images and text are tokenized jointly, and inference is run with deterministic decoding, low temperature, and a 10-token budget to restrict outputs to short semantic phrases. A lightweight post-processing step removes filler prefixes and normalizes phrasing to ensure metric consistency.

5) Problems Faced and Resolution

Natural-image bias and inconsistent output formatting were the primary challenges. Few-shot prompting reduced narrative drift, while fine tuning corrected visual-semantic alignment. Post-processing further standardized answer structure, improving stability across the evaluation set.

TABLE IV: SEMANTIC VQA PERFORMANCE ACROSS MODEL VARIANTS

Model	Setting	BERT-BLEU ₄
Qwen3-VL-8B-Instruct	Zero-shot Prompting	0.65
Qwen2-VL-7B-Instruct	Few-shot Prompting	0.70
Qwen2.5-VL-7B-Instruct-bnb-4bit	SFT (5k RS Samples)	0.85

D. Binary VQA

Binary Visual Question Answering (VQA) in Earth observation involves determining the presence, absence, or validity

of specific semantic or geometric attributes within a satellite image. Unlike captioning, which describes a scene holistically, binary VQA requires sharp discrimination of land-use patterns, object boundaries, and structural cues under varying illumination, scale, and sensor conditions.

1) Metric Evaluation

Binary predictions are evaluated using an exact-match criterion:

$$S_{\text{binary}} = \begin{cases} 1, & \text{if response}_{\text{pred}} = \text{response}_{\text{gr}}, \\ 0, & \text{otherwise.} \end{cases}$$

Any deviation from the canonical “yes” or “no” response is penalized, making this a strict test of both semantic precision and instruction adherence.

2) Difficulties

General-purpose VLMs frequently misinterpret industrial or agricultural structures due to natural-image priors. They also tend to produce unconstrained text instead of strict binary outputs. Reliable performance therefore requires domain-specific tuning, fixed response templates, and preprocessing strategies that minimize drift.

3) Testing and Performance

Zero-shot prompting produced inconsistent results, particularly on cluttered industrial scenes or small-object scenarios. Few-shot prompting stabilized formatting but did not fully correct semantic errors. Supervised fine tuning (SFT) on remote-sensing binary labels yielded the largest gains: the 4-bit Qwen2.5-VL-7B-Instruct model achieved a score of **0.915**, compared to 0.46 in the few-shot setting. These improvements reflect the importance of domain supervision for fine-grained yes/no decisions.

4) Experimental Setup

The binary VQA module uses a 4-bit Qwen2.5-VL-7B-Instruct model loaded via Unsloth’s FastVisionModel. The system resolves the appropriate adapter directory, applies inference optimizations, and wraps each query in a fixed instruction template restricting outputs to “yes” or “no.” Images and text are jointly tokenized, and inference is executed deterministically (no sampling, four-token limit, zero temperature) to avoid free-form responses. A lightweight post-processing step maps all variants of “yes” or “no” to canonical labels, ensuring compatibility with exact-match scoring.

5) Problems Faced and Resolution

Challenges included missing adapter checkpoints, inconsistent load paths, and slow repeated image tokenization. More critically, unconstrained model outputs harmed exact-match accuracy. These issues were resolved through unified loading utilities, caching, inference optimizations, and strict output

templates. The resulting pipeline is stable, efficient, and well aligned with binary evaluation requirements.

TABLE V: BINARY VQA PERFORMANCE ACROSS MODEL VARIANTS

Model	Technique Applied	Score
Qwen2-VL-2B-Instruct	Zero-shot + Prompting	0.54
Qwen2-VL-7B-Instruct	Zero-shot + Prompting	0.86
Qwen2.5-VL-2B-Instruct	Two-shot + Prompting	0.66
Llava-OneVision-Qwen2-7B	Zero-shot + Prompting	0.43
Qwen2.5-VL-7B-Instruct-bnb-4bit	SFT + Prompting	0.915

E. Grounding

Grounding requires the system to identify the exact region in an image that corresponds to a textual query. Unlike semantic VQA, which targets category-level meaning, grounding demands precise spatial alignment between language and visual structure. This is particularly difficult in remote-sensing imagery, where objects are small, spatial cues are subtle, and general-purpose VLMs lack reliable spatial understanding. To address these limitations, our approach relies heavily on SAM3 for mask generation, using the VLM primarily as a query parser rather than a spatial reasoner. The workflow is shown in Fig. 5.

1) Pipeline Overview

The process begins by passing the query and image through Qwen2.5-VL to extract four elements: the subject to ground, any reference objects, spatial relations (e.g., “left of”), and ordinal or size constraints. Because VLMs perform poorly on spatial reasoning in RS imagery, these relationships are converted into explicit coordinate-zone constraints rather than relying on the model’s internal perception.

SAM3 then performs two segmentation passes: one to generate candidate masks for the subject and another for reference objects. SAM3 is crucial here—it reliably identifies small, low-contrast structures that VLMs routinely misinterpret. Low-confidence masks are removed, and the remaining masks receive simple geometric descriptors such as relative size.

A lightweight VLM refinement loop evaluates each mask using the textual constraints, but only as a weak semantic signal. The core spatial decision-making is performed by a deterministic module that checks how well each mask matches

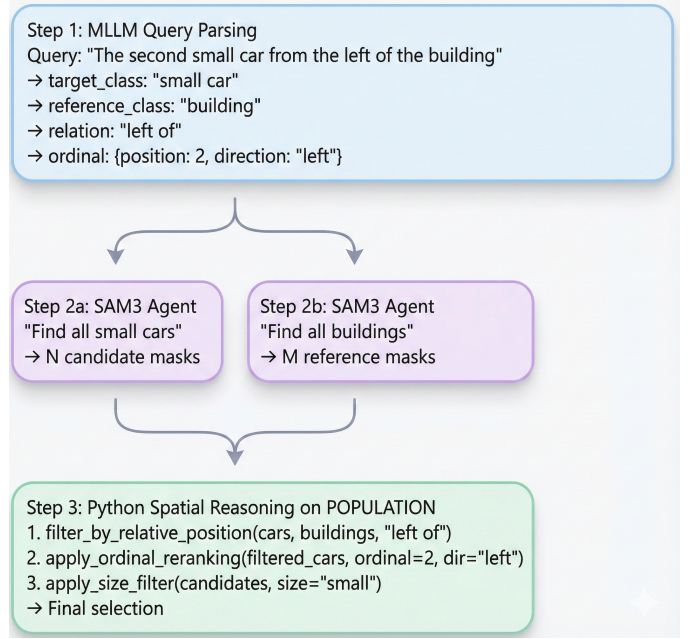


Fig. 5: Grounding workflow integrating VLM query parsing, SAM3 segmentation, spatial constraint reasoning, and final mask selection.

the relational constraints through geometric scoring. The final confidence for each mask is computed as:

$$\text{confidence} = 0.3 \cdot \text{VLM_decision} + 0.7 \cdot \text{spatial_score},$$

placing greater weight on SAM3-based spatial consistency. The selected masks are then converted into oriented bounding boxes.

Fig. 6 illustrates this process. SAM3 generates multiple airplane-like masks, but the deterministic spatial module selects the leftmost aircraft, addressing a failure mode common in VLM-only grounding systems.

2) Metric Evaluation

Grounding accuracy is evaluated using a multiplicative score that combines spatial alignment and count correctness. Let MeanIoU be the average IoU between predicted and reference oriented bounding boxes, and let CP be the count penalty:

$$S_{\text{grounding}} = CP \times \text{MeanIoU}.$$

The count penalty is:

$$CP = \exp(-\alpha \cdot |N_{\text{pred}} - N_{\text{ref}}|),$$

where N_{pred} and N_{ref} represent predicted and reference counts, with $\alpha = 2.5$. This formulation penalizes models that either miscount instances or misalign the spatial footprint.



Fig. 6: Example grounding output for the query: “Locate and return oriented bounding boxes for the left airplane.” SAM₃ produces candidate masks; spatial reasoning selects the correct aircraft.

3) Testing and Performance

Experiments on the VRSBench referring-expression split reveal the limitations of VLMs when used alone: zero-shot grounding frequently misinterpreted spatial cues, especially for elongated structures or closely packed objects. Introducing SAM₃ corrected these failures by providing reliable candidate masks, and deterministic spatial reasoning further stabilized performance on ordinal, relational, and region-based queries. The final hybrid system consistently identified correct instances even in challenging scenes with occlusions, shadows, and visually similar objects—conditions where VLMs typically struggle.

F. Dataset

All experiments in this work are conducted using VRSBench, a large-scale remote-sensing vision–language benchmark containing high-resolution imagery paired with captions, referring expressions, grounding annotations, and both numeric and semantic VQA labels [23]. The dataset covers diverse urban, industrial, agricultural, and airport scenes, making it well suited for evaluating multimodal reasoning under challenging remote-sensing conditions.

To ensure fair comparison and strict adherence to the official evaluation protocol, all metrics reported in this paper—captioning, semantic VQA, numeric VQA, and grounding—are computed solely on the VRSBench validation split. No images or annotations from this split are used during training or fine tuning. This separation prevents data leakage

and ensures that all results reflect true model generalization rather than memorization.

VRSBench’s diversity and task breadth allow us to evaluate each module of the GeoNLI pipeline under consistent conditions, from scene-level captioning to fine-grained object localization. Additional details on the dataset can be found at: <https://vrsbench.github.io/>.

IV. Conclusion

This work introduced a unified natural-language interface for remote sensing through the GeoNLI prototype, integrating captioning, numeric reasoning, semantic interpretation, and grounding within a single multimodal framework. The system combines parameter-efficient fine tuning with SAM₃-based spatial localization, addressing key weaknesses of general-purpose VLMs—particularly their limited spatial understanding and natural-image bias.

Across all tasks, supervised fine tuning and metric-aware prompting produced consistent gains. Captioning quality improved through concise, domain-aligned instruction design; numeric reasoning benefited from a hybrid routing strategy that uses a fine-tuned VLM for counting and SAM₃ grounding for geometric queries; and semantic VQA showed strong improvements when trained on curated remote-sensing samples. Grounding performance, in particular, highlighted the importance of SAM₃, whose reliable mask generation compensates for the spatial reasoning limitations of vision–language models.

Despite relying on lightweight quantized models, the system maintained stability and low inference cost, demonstrating that domain adaptation—not model scale—is the key driver of performance in remote-sensing applications. The resulting pipeline is capable of robust small-object detection, relational reasoning, and concise scene interpretation, making it suitable for real-world decision-support contexts such as infrastructure monitoring and rapid assessment.

Future work will explore sensor-generalization, multi-temporal reasoning, and tighter integration with large-scale remote-sensing workflows to further validate the system’s applicability and broaden its deployment potential.

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